

FT8-Contest-Helper-3.0.x User Guide

The FT Contest Helper (hereinafter “Helper”) program was developed as a macro extension using Quick-Macros (© Gintaras Didzgalvis). Unlike the FT8 Helper, the FT Contest Helper **cannot act as an “FT8 robot.”** The operator must manually approve each QSO to ensure compliance with the requirements for legal use in contests. This naturally raises the question: **Why use the FT Contest Helper at all?**

The Helper handles time-sensitive tasks that are very helpful before Allam in S&P mode.

- Find and initiate the next QSO with the best chances and the most points.
- Stop calling after the set maximum number of attempts.
- Prioritize by multiplier or distance, depending on the contest type.
- All contests implemented in WJST-X are supported.
- To test the FT Contest Helper, regular QSOs are also supported, but they offer much less convenience than the FT8 Helper (no band hopping, etc.).
- S&P, RUN (CQ) and mixed SP/CQ operation are supported.
- The helper analyzes and stores all messages in the background, so stations with new multipliers can be called preferentially even if they are only transmitting RR73 or 73.
- Stations with more than 2 unsuccessful QSO attempts are blocked for a certain period of time.

The Helper program interprets the data received by WSJTX and acts intelligently according to various selectable strategies. The Helper does not require QuickMacros to run. It controls WSJTX via virtual mouse and keyboard commands. While the Helper is running, you should not use your computer for other applications, as the mouse focus will be taken away from the FT8 Helper every 7.5 or 15 seconds, depending on the FT4/FT8 mode.

Please note that the WSJTX version number (here 3.0.3) must match the Helper’s version number. The number behind “3.0.2- ” is the Helper’s sequential version number.

This version is a free “trial version” and can be used without restrictions until the end of 2026.

QSO Operation:

Before starting, you must configure a few settings in FT-Contest-Helper and WSJT-X!

After starting, the Start/Stop indicator changes from “STOP” to “CQ” or “SP” and the display turns “green.” When a QSO begins, the mode indicator turns “yellow” and displays “START-next.” From that moment on, you can start the next QSO simply by pressing the START button. The display then turns “green.”

If the station does not respond, the QSO is terminated after the set number of retries. If a QSO with a station is terminated 3 times, that station is blocked (postponed) for a certain period of time.

Contest-Helper 3.0.2-2 by DG5LP

Own call / grid /

Mode Select Band ms

☒ Hold Tx

FT-mode Contest Strategy ☒ 73=CQ

☐ Skip Tx1

17:57:37: Running CQ/SP Mode stopped
 17:57:33: Waiting to stop CQ/SP-Mode
 17:56:37: Station blocked for a while: WF8E
 17:56:37: QSO logged with: WF8E
 17:56:22: QSO started with WF8E
 17:55:37: QSO broken with: WF8E (Timeout)
 17:54:22: QSO started with WF8E
 17:53:37: QSO logged with: CT7ALI
 17:53:07: QSO started with CT7ALI
 17:52:22: QSO logged with: UY0HY
 17:51:52: QSO started with UY0HY
 17:51:37: QSO broken with: 9A60CBM (No answer)
 17:50:52: QSO started with 9A60CBM
 17:50:37: QSO logged with: CT7AUT
 17:50:22: QSO started with CT7AUT
 17:50:22: QSO broken with: CT7ALI (Replies to other)
 17:49:37: QSO started with CT7ALI
 17:49:22: QSO broken with: CT7AUT (Replies to other)
 17:48:52: QSO started with CT7AUT
 17:48:07: QSO broken with: WF8E (Timeout)

Worked/Multipliers	Stations/new Multipliers	CQ on QRG
CT7ALI CT	9A60CBM 9A	9A60CBM 805
CT7AUT CT	CT1FPQ CT	YO4NT 2337
EA2AA EA	EA1ESW EA	
JA2AA JA	EA1FAS EA	
LA2AA LA	EA3HYJ EA	
M7IDD G	EA5HKZ EA	
M9POB G	EA7O EA	
R2PX UA	F1TXI F	
SP0LPR SP	R3DIU UA	
UA2AA UA2	RZ2A UA	
UY0HY UR	SV3TPT SV	
WF8E K	YO4NT YO	

QSOs:

Main Window of the FT-Contest-Helper

The Main Window - Data Fields

All events, such as start, break, and logging of QSOs, are recorded. Band changes, FT modes, and contests are also logged.

Worked/Multiplier Field:

The data displayed depends on the selected contest:

- For contests without multipliers: Regular QSO, ARRL-DIGI, and ARRL-FD
The list of logged stations with their DXCC designators, e.g., W1AA K

- For contests with multipliers: WW-DIGI, FT-RU, NA-VHF, and EU-VHF

The list of multipliers worked, along with the number of QSOs with each multiplier, e.g.:

- WW-DIGI: **JN 5** i.e. 5 QSOs with JN
- FT-RU: **DL 25 (DX), LAX 2 (US), or QC 3 (VE)**
- NA-VHF: **JN48 20**
- EU-VHF: **JN48FX 5**

This list is saved and loaded upon restart. The list is deleted when changing the contest (after a confirmation prompt).

Stations/new Multipliers Field:

The data displayed depends on the selected contest:

- For contests without a multiplier:

The list of stations that have been logged but not yet worked, along with their DXCC identifiers. This list is not saved and is deleted when you change bands or FT modes.

- For contests with a multiplier:

The list of stations that have not yet been worked and that have a new multiplier. The stations on this list are given priority. When a multiplier is worked, all stations with that multiplier are deleted.

This list is saved and restored when the helper is restarted. The list is deleted only when switching contests (after a confirmation prompt).

CQ-on QRG Field:

The list of received CQ calls, along with the frequency and station callsigns. If a station on this frequency later sends a 73 or RR73, the helper can call that station. If a multiplier was also received from the station and the station appears on the "Stations/New Multiplier" list, it will be called with higher priority.

The Main Window - Controls

Own call / grid:

Your own call sign and QRA locator, must be entered into the configuration using the "Edit config" dialog.

Edit config:

Opens the configuration dialog of the Helper. The configuration data is stored in the "config.txt" file located in the installation folder. See more details below in the text.

About:

Displays the current version number and copyright information.

Mode Select:

Use "Mode-Sel" to select the desired operating mode of operation: S/P CQ, or MIX. The specific functions of each mode are described below.

S/P-Mode:

In "Search and Pounce" mode, the Helper selects stations calling CQ. Which stations are called first, depends on the configured "Auto-Strategy" (see below). When the function 73=CQ is enabled, stations sending "73" or "RR73" will also be called, provided that the Helper has previously received "CQ" from that station in the background.

CQ-Mode:

In CQ mode, WSJT-X continuously transmits CQ calls. It is important to note that the helper decides which station to call back. If multiple stations call simultaneously, the station is called back according to priority. Stations with a new multiplier always take precedence. The helper starts CQ mode with "Hold-Tx" enabled. Please note that WSJT-X should not respond to the CQ calls, so CQ must be set to "None."

MIX- Mode:

In MIX mode, the Helper calls CQ and waits for responses. If no one responds, the Helper calls stations that are calling CQ, in accordance with the configured Auto Strategy. Upon completion of the QSO, it resumes calling CQ. If, once again, no one responds, it searches for a CQ-calling station again.

Note: The FT Contest Helper uses WSJT-X's auto-sequencing, but the automatic response to CQs must be set to "CQ-None."

Band:

Selects the desired band. The frequencies required for the contest must be preset in WSJT-X. Starting with version 3 of WSJT-X, band switching is done using buttons instead of a drop-down menu. To enable this, check the "Band Buttons" box in the "View" menu of WSJT-X.

START / STOP and Operation Indicator:

As described above, to comply with contest rules, all QSOs must be authorized by pressing the "START" button.

After starting, the mode indicator changes from "STOP" to "CQ" or "SP" and the display turns "green." When a QSO begins, the mode indicator turns "yellow" and displays "START-next." From that moment on, you can start the next QSO simply by pressing the START button. The display then turns "green."

Pressing the "STOP" button initiates the shutdown process, but the system waits until the current QSO is complete. The operation indicator turns "yellow" and displays "Wait-Down." Pressing the "STOP" button again immediately terminates all activities.

FT Mode:

In WSJT-X Version 3, the selection of FT modes was changed from drop-down menu to individual buttons. The Helper uses these new buttons to switch between FT-4 and FT-8. The FT mode can only be changed while the program is paused.

Contest:

The desired contest is set here. The strategy for prioritizing stations is selected automatically. To adapt to future contests with different rules, you can change the "Strategy."

Contest	Strategy	Exchange	Multiplier
Regular QSO	Best S/N oder Prefer-DX	S/N Report	Kein Multiplier
WW-DIGI	Prefer DX	2-Char. QRA-Lokator	2-Char. QRA-Locator
FT-RU	DXCC & US/VE States	599 0001	DXCC & US/VE States
ARRL-DIGI	Prefer DX	4-char QRA-Locator	Kein Multiplier
NA-VHF	Best S/N	4-char. QRA-Lokator	4-char. QRA-Locator
EU-VHF	Prefer DX	5990001 + 6-char QRA.	6-char QRA-Locator
ARRL-FD	Best S/N	FD-Exchange oder DX	Kein Multiplier

Strategy: Best S/N:

The station calling CQ or transmitting 73/RR73 with the highest priority is called. If multiple stations with the same priority are received, the station with the better S/N ratio is called. Priorities are also taken into account for incoming calls.

The Helper distinguishes between 5 priority levels, with "Wanted" being the highest priority:

- **"Wanted" priority:** The station matches the "Wanted" entries in the Helper's configuration.

- **“High” priority:** The station is a new multiplier, or is highlighted in WSJT-X as “New Call” or higher.
- **“Normal” priority:** The station is not a new multiplier and is highlighted in WSJT as “New Call on Band.”
- **“Low” priority:** After two unsuccessful QSO attempts, the priority is set to “Low.”
- **“Blocked” priority:** After three unsuccessful QSO attempts, the station is blocked for a while. Stations are also blocked if they do not respond with the correct exchange, e.g., the station transmits S/N when a QRA locator or serial number is expected.

Strategy Prefer-DX:

Priorities are assigned in a manner similar to “Best S/N.” If multiple stations have the same priority, the station that is farthest away is contacted.

Decode Delay (Lag time):

In most cases WSJTX cannot decode all messages within 15/7,5 sec. This value indicates how long WSJTX requires for decoding the messages during the subsequent cycle. The decoding depth in WSJTX should be configured such that this value remains below 500ms. If it exceeds 1000 msec, there is a risk that transmitted messages may not be decoded by the receiving station.

Hold-TX:

In CQ and Mix modes, the Helper selects “Hold-TX”; in S/P mode, “Hold-TX” is turned off but can be switched on as needed. “Hold-TX” is generally recommended, but it has disadvantages in S/P mode.

73=CQ:

If this function is selected in S/P mode, the Helper will call not only stations with CQ, but also stations sending RR73 or 73, provided that the Helper has previously received a “CQ” from that station in the background.

If you set “Hold-Tx-Freq” manually in S/P mode and you are on a free frequency, the Helper calls stations even if no CQ was previously received.

Skip Tx-1:

If Skip Tx-1 is selected during regular QSO Mode, the Helper calls stations by sending an S/N report instead of the QRA locator. This can help shorten the QSO slightly. This setting is ignored for all contest modes. During contest operation, stations are always called using their CALL or report, in accordance with the contest rules.

Footer Line:

1. The number of QSOs in the contest. The counter is reset when the contest changes.
2. The station currently being worked. The field turns green when the station has been logged and red when the QSO has been aborted.
3. Status/progress of the QSO. E.g., WAIT-CALL, WAIT-REP, WAIT-RREP, WAIT-RR73, etc. The number of attempts is also displayed.
4. EXIT: Exits the program.

The „Edit Config“ Dialog

Configuration Dialog

My-Call My Grid My Locations

Macro Delay (if required) msec Minimum S/N level accepted

Max. audio frequency Nr of CQs in MIX before S/P

Max. CALL repeats Max. S/P QSOs in Mix-mode

Max. repeats in QSO Logging level

Max repeats at QSO end Excluded Calls

Minimum DX Km

Wanted Grids

Wanted DXCCs

- My-Call ; your station`s call-sign
- My-Grid ; the four character QRA locator (grid) of your own station
- MY-Locations ; Your own country and continent. In regular QSO mode, this allows the Helper to respond only to CQ calls that are directed at you. In contest mode, you can enter non-standard contest names here. Standard names, such as "TEST," "WW," and "RU," are recognized automatically.
- Macro Delay ; Some computers require a value between 10 and 50 here
- Max. audio freq. ; Stations calling above the specified frequency will be ignored.
- Max. CALL repeats ; Maximum number of call repeats when calling stations. If this number is reached and the station does not respond, the QSO is terminated.
- Max. repeats in QSO ; Maximum number of repetitions when exchanging reports. If this number is reached and the station does not respond, the QSO is terminated.
- Max. repeats at QSO-end ; Maximum number of repeats at the end of the QSO, after the QSO has been logged
- Min. DX distance ; The minimum DX distance is evaluated only in regular QSO-Mode. This prevents the helper from responding to CQ-DX calls if the distance to the station is less than the specified value, typically 2000 Km.
- Min. S/N level ; Stations with an S/N value below this value are ignored
- Nr. of CQs in MIX ; Maximum Number of consecutive CQ calls transmitted in MIX mode before the helper switches to the RUN-SP phase.
- Nr. of S/P QSOs in Mix ; Maximum number of S/P QSOs in MIX-Mode during the RUN-S/P phase before the helper switches back to the RUN-CQ phase.
- Logging level ; Log is written into the daily logfile "\\log\\log-<YYMMDD.txt" in the installation directory;
 - Logging level: NONE ; No logging file written
 - Logging Level: LOG ; Only the content of the log-window of the FT8-Helper is logged
 - Logging level: MSG ; Additionally, all received and transmitted messages are logged
 - Logging level: EXPERT ; Additionally, information for software debugging are logged

- Excluded Calls ; List of stations and prefixes you **don't want** to work. A "JA1" it applies only to call signs whose first 3 characters are "JA1". Invalid countries, such as "D0," can only be entered here.
- Wanted-Grids ; List of QRA locators, separated by spaces, that you want to work. Two or four characters are accepted.
- Wanted DXCCs ; List of DXCC countries and continents you would like to work. The DXCC prefixes specified in WSJTX are allowed, as well as the continents *EU, *AF, *AS, *NA, *SA, and *OC.

The Require Settings

Settings in Windows 10/11:

- The minimum required resolution is 1680 x 1050 pixels.
- Set the Windows system's screen scaling to 100%.
- Use only the default Windows window manager.
- Some remote programs may have unexpected effects on functionality.

Required settings in WSJTX

Main Window

- During contests, "CQ/73" must not be selected
- "Hold Tx Freq" is controlled by the helper, do not touch here.
- "Tx even/1st" should be changed occasionally during a contest.
- Band and FT8/FT4 are controlled by the helper, do not change here
- Auto Seq. is controlled by the helper. Do not change
- CQ: None should always be selected; the helper selects the stations themselves.
- Enable / Halt Tx is controlled by the helper, do not touch here
- The "Menu" checkbox must be checked. The helper will issue a warning if it is not.
- The sizes of WSJT-X and Wide-Graph set by the helper must not be changed.
- The split of WSJT-X between "Band Activity" and "Rx Frequency" should be approximately 50%-50%.
- The windows may be moved, but must always remain entirely on the desktop.
- The windows should not overlap, nor should any external software.

View Menu:

- "Use Dark Style" must not be enabled.
- "Band Buttons" must be enabled. The helper will warn you if it is not.

Filters Menu:

You can specify filters, but make sure not to filter out any messages that are important for the contest.

Mode Menu:

The FT8/FT4 setting should be configured via the Helper.

Decode Menu:

The settings in the "Decode" menu should be adjusted according to the performance of your PC. Monitor the "Decode Delay" display in the Helper: the value should not exceed 500 ms.

Settings in WSJT-X "General":

The screenshot shows the 'Settings' dialog box for WSJT-X, with the 'General' tab selected. The 'Station Details' section contains fields for 'My Call' (DG5LP), 'My Grid' (JN48f), an 'AutoGrid' checkbox, and an 'IARU Region' dropdown (All). Below this is a dropdown for 'Message generation for type 2 compound callsign holders' set to 'Full call in Tx3'. The 'Display' section has several checkboxes: 'Start new period decodes at top' (unchecked), 'Blank line between decoding periods' (checked), 'Show blank line with time stamp' (checked), 'Display distance in miles' (unchecked), 'Highlight DX Call in message' (checked), 'Tx messages to Rx frequency window' (checked), 'Highlight DX Grid in message' (unchecked), 'Show DXCC, grid, and worked-before status' (checked), 'Show principal prefix instead of country name' (checked), and 'Show DXCC for all messages' (unchecked). There are also buttons for 'Font...' and 'Decoded Text Font...'. The 'Behavior' section includes checkboxes for 'Monitor off at startup', 'Monitor returns to last used frequency', 'Double-click on call sets Tx enable', 'Disable Tx after sending 73' (checked), 'Calling CQ forces Call 1st', 'Alternate F1-F6 bindings', 'Enable VHF and submode features', 'Allow Tx frequency changes while transmitting', 'Decode after EME delay', 'Single decode', 'MSK144/Q65: Tx until 73 is received', 'Auto-open/close Astronomical data window', 'kHz entry without k (restart required)', 'Turn progress bar red while transmitting' (checked), 'CW ID after 73' (unchecked), 'ID Interval' (0), and 'Tx watchdog' (6 minutes). The 'Additional features' section has checkboxes for 'Map Grid locator to U.S. State', 'Show U.S. States in messages with grid', 'Clear DX Call after QSO', 'Show distance in messages with grid', 'Erase decodes after band change', 'Set Rx frequency to Tx frequency after QSO', 'Clear DX Grid after QSO', 'Show azimuth in messages with grid', 'Alternate Erase button behavior' (checked), 'Enable Wait features' (checked), 'Less button coloring (restart required)' (unchecked), and 'Align DXCC/distance/azimuth' (checked) with values 2 and 0. At the bottom are 'OK' and 'Cancel' buttons.

- The "Font" setting must be: MS Shell Dlg 2, Font style: Standard, Size: 8,
- The "Decoded Text Font" setting must be: Courier or Courier New, Font style: Standard, Size: 10,
- These values are the default settings after installing WSJT-X.

Settings in WSJT-X "Reporting":

"Log automatically" must be selected. All other checkboxes in the "Logging" section must remain unchecked.

Settings in "WSJT-X Frequencies":

Make sure that the correct frequencies required for the contest are programmed for both FT4 and FT8

FT8.Settings in WSJT-X "Colors":

- The "Highlighting" color "CQ in message" (green) must be selected.
- "Transmitted message" (yellow) should also be selected.
- "New Call on Band" must be selected.
- Other color markings can be selected as needed.
- "Highlight messages with 73 or RR73" should be enabled.
- The color scheme must not be changed. The helper only recognizes the original colors.

Correct settings of the Wide-Graph window:

- Bins/Pixel = 3
- Start 500 Hz
- All frequencies from 500 Hz to 2850 Hz must be visible.

Installation and Preparation Before First Use

The FT Contest Helper is “portable software” that does not require installation on Windows. It can be launched from any directory (including a USB drive).

Download the program (see the link at www.github.com/dg5lp) and copy it to any directory. Launch WSJT-X, then the FT Contest Helper. When you start it for the first time, your call sign and QRA locator will be blank. Open the “Edit Config” dialog and enter your personal information in the three fields: “My-Call,” “My Grid,” and “My Locations.” To save the configuration, don’t forget to click “OK.” The Contest Helper is now ready for use. Select “Contest” or “Regular QSO” and start “S/P,” “CQ,” or “MIX” mode.

Bug reports and observations to:

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